

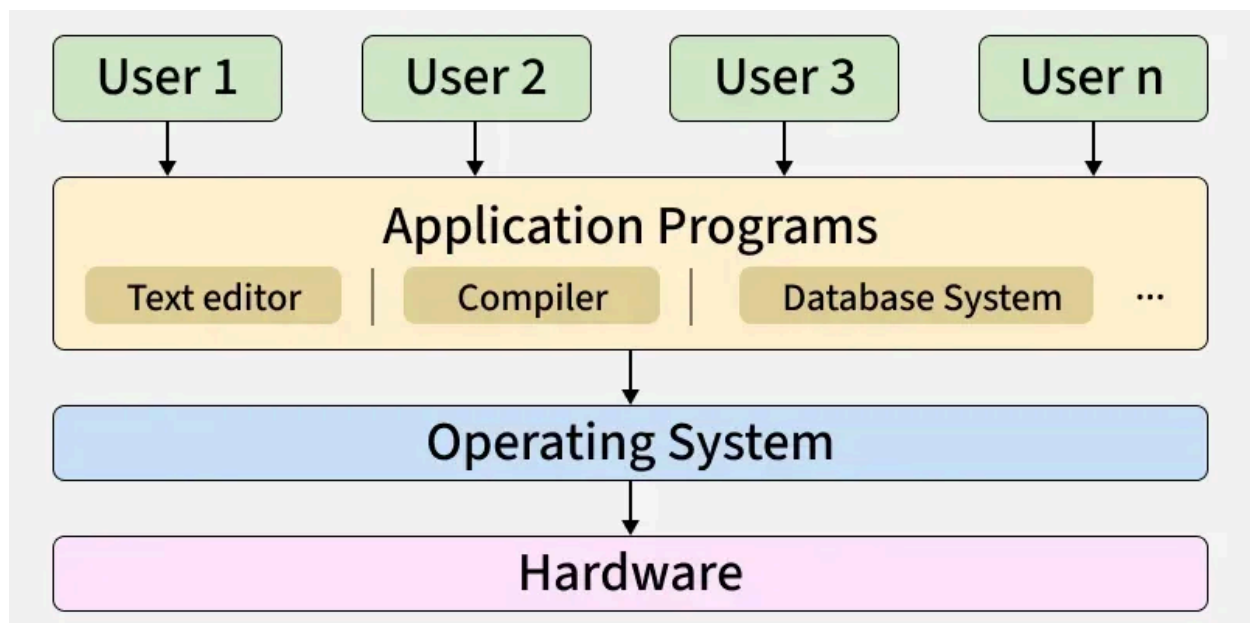


Operating System Tutorial

Last Updated : 27 Jan, 2026

An Operating System(OS) is a software that manages and handles hardware and software resources of a computing device.

- Manages computer resources such as CPU, memory, and files
- Acts as an interface between user and hardware
- Performs core functions like process, memory, and file management
- Organizes system resources similar to different departments in a government
- Examples include Linux, Unix, Windows 11, MS-DOS, Android, macOS, and iOS



Basics

This section introduces the fundamentals of operating systems.

1. [Introduction](#)
2. [Types of OS](#)
3. [Kernel in OS](#)
4. [System Call](#)
5. [System Initialization](#)

Process Scheduling

Focuses on process management and scheduling in operating systems.

1. [Process Introduction](#)
2. [Process Control Block](#)

3. [Process Table](#)
4. [Process Management Introduction](#)
5. [Process States](#)
6. [Process Scheduler](#)
7. [CPU Scheduling Algorithms](#)
8. [Preemptive vs Non-Preemptive](#)
9. [Dispatcher vs scheduler](#)
10. [Starvation and Aging](#)

Quiz: [CPU Scheduling](#)

Process Synchronization

Covers synchronization and IPC concepts.

1. [Inter Process Communication](#)
2. [Process Synchronization](#)
3. [Race Condition](#)
4. [Critical Section](#)
5. [Solutions to Process Synchronization Problems](#)
6. [Peterson's Algorithm](#)
7. [Dekker's algorithm](#)
8. [Bakery Algorithm](#)
9. [Hardware Based Solutions](#)
10. [Semaphores](#)
11. [Mutex vs. Semaphore](#)
12. [Monitors](#)
13. [Priority Inversion](#)
14. [Classical IPC Problems](#)

Quiz: [Process Synchronization in OS](#)

Deadlock

Explains deadlock in operating systems.

1. [Introduction](#)
2. [Deadlock Handling](#)
3. [Deadlock Prevention](#)
4. [Banker's Algorithm for Deadlock Avoidance](#)
5. [Detection And Recovery](#)
6. [Starvation, and Livelock](#)
7. [Resource Allocation Graph \(RAG\)](#)
8. [Methods of resource allocation](#)
9. [Program for Deadlock free condition](#)

Quiz: [Deadlock](#)

Multithreading

Focuses on multithreading concepts in operating systems.

1. [Operating System | Thread](#)
2. [User Level Vs Kernel Level threads](#)
3. [Process-based and Thread-based Multitasking](#)
4. [Multi threading models](#)
5. [Benefits of Multithreading](#)

Quiz: [Multithreading](#)

Memory Management

Introduces the fundamentals of memory management in operating systems.

1. Basics

- [Introduction to memory and memory units](#)
- [Memory Management in Operating System](#)
- [Logical and Physical Address in Operating System](#)

2. Contiguous Allocation

- [Implementation of Contiguous Memory Management](#)
- [Internal Fragmentation](#)
- [External Fragmentation](#)
- [Program for Next Fit algorithm in Memory Management](#)
- [Buddy System: Memory allocation technique](#)

3. Non-Contiguous Allocation

- [Non-Contiguous Allocation in Operating System](#)
- [Paging](#)
- [Page Table Entries in Page Table](#)
- [Segmentation](#)
- [Segmentation with Paging](#)

4. Advanced Memory Concepts

- [Overlays in Memory Management](#)
- [Virtual Memory](#)
- [Demand Paging](#)
- [Page Fault Handling](#)
- [Swap Space](#)

5. Page Replacement & Thrashing

- [Page Replacement Algorithms](#)
- [Belady's Anomaly](#)
- [Second Chance \(or Clock\) Page Replacement Policy](#)
- [Techniques to handle Thrashing](#)

- [Working Set in Paging](#)

6. Kernel & System-Level Concepts

- [Allocating kernel memory \(buddy system and slab system\)](#)
- [Memory Interleaving](#)
- [Operating system based Virtualization](#)

Quiz: [Memory Management](#)

Disk Management

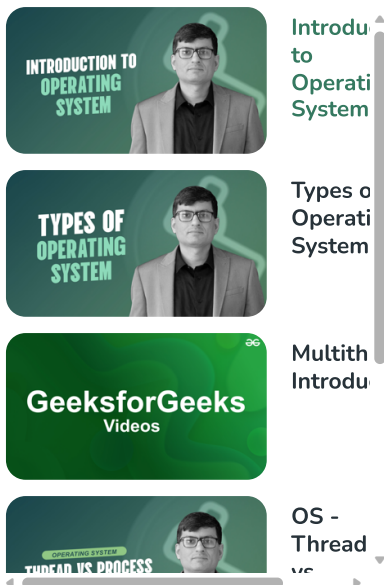
Explains disk and file management in operating systems.

1. [File Systems](#)
2. [Unix File System](#)
3. [File Directory | Path Name](#)
4. [Structures of Directory](#)
5. [File Allocation Methods](#)
6. [File Access Methods](#)
7. [Secondary memory](#)
8. [Secondary memory – Hard disk drive](#)
9. [Disk Scheduling Algorithms](#)
10. [Program for SSTF disk scheduling algorithm](#)
11. [What exactly Spooling is all about?](#)
12. [Spooling vs Buffering](#)
13. [Free space management](#)

Quiz: [Input Output Systems](#)

Important Links

1. [Last Minute Notes \(LMNs\) | Operating Systems](#)
2. [Commonly Asked Operating Systems Interview Questions](#)
3. ['Quizzes' on Operating Systems !](#)
4. [GATE CSE Previous Year Questions](#)



Introduction to Operating System

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